
Temporal Redundancy Breaks JEPA on Satellite Image Time Series

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Abstract

1 Earth observation data is natively a time series of the same location, yet most
2 self-supervised objectives for satellite imagery treat acquisitions independently.
3 Predicting the *future* latent state of a location from its past should be the better-
4 matched objective. We show that a direct port of the JEPA objective to Sentinel-
5 2 time series **collapses**: because consecutive acquisitions of a parcel are near-
6 identical, future-latent prediction is solvable by emitting a constant (loss $\rightarrow 0$,
7 per-dimension std 0.04, effective rank 2.4 of 512). Adding variance–covariance
8 regularization on the context embedding removes the degenerate solution. With it,
9 causal temporal prediction outperforms spatial masking by +6.0 mIoU on PASTIS
10 crop segmentation under a frozen probe (22.3 vs. 16.2, paired t -test $p = 0.041$,
11 3 seeds), and the margin widens from +37% at full supervision to +100% at 1%
12 labels. Probing frozen spatial features shows the objective makes the representation
13 phenology-aware: acquisition month decodes at 61.3% vs. 46.3% (chance 8.3%).

14 1 Introduction

15 Contributions.

- 16 • We identify and measure a failure mode specific to satellite image time series: temporal
17 redundancy makes latent future-prediction degenerate, and we quantify the collapse via
18 effective rank and per-dimension variance.
- 19 • Under a matched-backbone, matched-budget protocol, causal temporal prediction out-
20 performs spatial masking and reconstruction/contrastive baselines on PASTIS, with the
21 advantage growing as labels shrink.
- 22 • We give direct evidence for the mechanism: the temporal objective makes *single-frame*
23 spatial features season-aware, which is the crop-discriminative signal.

24 2 Method

25 **Setup.** Given $X \in \mathbb{R}^{T \times 10 \times 128 \times 128}$ with acquisition days $d \in [1, 366]^T$, the encoder maps each
26 frame to $N=256$ tokens of width $D=512$ ($P=8$ patches). A temporal transformer aggregates frames
27 $1:t$ causally; a narrow predictor ($D_p=384$) maps the context to the EMA-target embedding of frame
28 $t+\Delta$. Day-of-year enters only as a periodic encoding; chronological order is the acquisition index.

Figure 1: Effective rank of the context embedding over pretraining on PASTIS. Without variance–covariance regularization the representation collapses to rank ≈ 2.4 of 512 while the prediction loss goes to zero — the degenerate solution admitted by near-identical consecutive acquisitions.

Table 1: PASTIS crop segmentation, frozen encoder, validation fold. Mean $\pm$ std over 3 seeds; paired t -test against temporal $\Delta=1$. All methods use an identical backbone and effective batch 192 for 100 epochs.

Objective	conv mIoU	p vs. temporal	parcel k -NN
Raw spectral features	FLOOR-TBD	FLOOR-TBD	FLOOR-TBD
Random-init encoder	FLOOR-TBD	FLOOR-TBD	FLOOR-TBD
MAE	TBD	0.001–0.009	54.4
SimCLR	TBD	0.001–0.009	54.6
BYOL	TBD	0.001–0.009	62.7
Spatial JEPA	16.2	0.041	58.7
Spatial JEPA (compute-matched, $3.5 \times$ epochs)	15.8 ± 1.2	0.036	–
Temporal JEPA ($\Delta=1$)	22.3 ± 1.8	—	65.5

29 3 Temporal redundancy induces collapse

30 4 Experiments

31 5 Results

32 6 Why it wins: the representation becomes season-aware

33 7 Limitations

34 8 Conclusion

Figure 2: Test-fold mIoU at 1/5/10/100% label fractions. Temporal: 9.2/13.1/15.9; spatial: 4.6/6.9/9.5. The advantage grows from +37% at full supervision to +100% at 1% labels.